Supplementary Material for cfRNA-BrainTrace

Supplementary Methods

S1 Formal validation route

All validation analyses followed the same formal route used by the software. A query profile was first represented in logCPM- or logTPM-compatible expression space and then projected into Bo2023-like VSD space for Network scoring only. The top three Networks formed the candidate beam. Resolution-group and exact-region candidates were then reranked within that beam using local logCPM-compatible expression. This design separates broad candidate generation from fine regional interpretation.

S2 Internal validation design

Internal validation used the Bo2023 macaque brain reference. Two settings were evaluated. Leave-one-sample-out validation tested whether the route could recover the held-out sample label when other samples from the reference remained available. Leave-one-monkey-out validation held out all samples from one animal at a time and therefore tested donor-level generalization. Metrics were reported separately for Network, resolution-group and exact-region levels. Top1, Top3 and median true-rank were used to distinguish exact calls from candidate-list recovery.

S3 External validation design

External validation was limited by the label resolution available in each dataset. AHBA human brain RNA-seq was used for mapped-label validation because its anatomical labels could be harmonized to Network, resolution-group and a subset of exact-region labels. TCGA/BraTS glioma tissue RNA-seq with MRI-derived labels was used only for coarse anatomical consistency because its truth labels are human imaging labels, not Bo2023 macaque exact-region identifiers. GSE189919 was used to test whether an external matrix could be projected into the model gene space; it was not used for accuracy estimation because patient-level anatomical truth was unavailable.

S4 Label harmonization and allowed conclusions

Label harmonization was performed only to the level supported by each dataset. AHBA anatomical labels were mapped to the macaque-derived Network, resolution-group and exact-region hierarchy where a supported mapping existed, and results are interpreted as mapped-label transfer rather than direct anatomical equivalence. TCGA/BraTS labels were derived from human MRI/tumour context and therefore support only coarse tumour-tissue anatomical consistency. Biofluid datasets without patient-level anatomical truth were not used for localization accuracy and are reported only as projection-feasibility or transfer stress tests.

S5 Model artifacts and projected-VSD projector

The repository contains lightweight model artifacts in data/models/, including Network model files, region-resolution dictionaries, route metadata and the reference-fitted projector bo2023_reference_projector_linear_full.npz. The projector stores gene-wise slope, intercept and clipping parameters fitted from reference data. At inference time, a single query profile can be mapped into Bo2023-like projected-VSD space using these fixed parameters without target-cohort labels or target-cohort distribution fitting. Projected VSD is used only for Network Top3 beam generation; resolution-group and exploratory exact-region reranking use logCPM-compatible local expression within that beam.

S6 Example input/output

Synthetic public example files are provided in submission_ready_assets/example_io/ as part of the v0.1.6 public submission release. They document accepted input columns, Network ranked candidates, three-tier JSON output, resolution-group ranked candidates, exact-region ranked candidates and the local generation script. The example files are format and reproducibility aids only; they are not biological validation samples.

S6a Software availability

The manuscript-associated software release is v0.1.6. The public GitHub repository is available at https://github.com/wz7717/cfrna-brain-tracing, and the archived release is cited in the manuscript using the Zenodo DOI.

S7 Model development and locked evaluation timeline

The production route was fixed before generating the final submission validation tables. External AHBA, TCGA/BraTS and biofluid analyses were not used to select the final route; they were used only for mapped-label transfer evaluation, coarse tumour-tissue consistency assessment and biofluid transfer stress testing. Development comparisons that used the same internal validation framework are reported as development evidence rather than independent confirmation.

S8 Figures and tables

Supplementary Figure S1 provides the main route and validation summary artwork. Supplementary Tables S1-S6 provide the internal validation design, internal validation results, external validation design, external validation results, figure/table index and claim-boundary summary. Together, these materials document both how the route was tested and what result each validation setting supports.

Supplementary Results

S1 Internal route selection

The first internal analysis evaluated Network-level candidate generation. Projected VSD achieved Network Top1/Top3 of 58.00%/91.58% in LOSO and 53.72%/91.33% in LOMO, exceeding logCPM baseline and native VSD at Network Top3. Direct exact-region scoring was lower and less stable, especially in LOMO. These results motivated the use of projected VSD for broad Network beam generation and logCPM-compatible expression for downstream local reranking.

S2 Formal internal three-tier validation

In the complete LOSO validation, the formal route achieved Network, resolution-group and exact-region Top3 values of 92.38%, 72.36% and 45.33%. In complete LOMO validation, the corresponding Top3 values were 91.21%, 69.09% and 42.36%. Median true-rank increased from Network to exact-region levels, consistent with decreasing anatomical certainty at finer resolution. Resolution group is therefore the preferred region-level endpoint, while exact-region output is retained as a candidate ranking.

S3 External validation results

In AHBA, the hybrid route achieved Network Top1/Top3 of 74.68%/94.42%, resolution-group Top1/Top3 of 36.26%/67.03% and exact-region Top1/Top3 of 24.18%/42.86%. Hybrid exact Top3 exceeded logCPM baseline (30.77%) and projected-VSD-only scoring (29.67%). In TCGA/BraTS, hybrid Network Top3 was 40.00% and broad-anatomy Top3 was 64.62%, supporting only coarse anatomical consistency. GSE189919 overlapped 15,622/21,668 projector genes, corresponding to 72.10% gene-space coverage, supporting projection feasibility rather than source-localization accuracy.